

Flutter Navigation & Reusable Widgets

Implementation Report — Navigation, Custom Widgets, ListView, GridView and Bottom Navigation

Project	My First Flutter Project
IDE	Visual Studio Code
Platform	Flutter / Android
Task	Learn Navigation and Reusable Widgets

Project Implementation Report

1. Project Overview

This task extends the Flutter application by introducing navigation, reusable widgets, organized folders, product cards, ListView and GridView layouts, and a BottomNavigationBar. The screens are connected through Flutter navigation and named routes, while repeated UI elements are moved into reusable widget files.

2. Tasks Completed

- Navigation between Splash, Login, Sign Up and Home screens.
- Named routes configured for application navigation.
- Reusable Custom Button widget created.
- Reusable Custom TextField widget created.
- Reusable Product Card widget created.
- BottomNavigationBar added with Home, Profile and Settings sections.
- ListView and GridView implemented for displaying products.
- Project organized into screens and widgets folders.

3. Topics Covered

Navigator	Routes
Custom Widgets	ListView
GridView	BottomNavigationBar

4. Navigation & Routes

The application uses Flutter Navigator with named routes. The main route configuration connects the Splash Screen, Login Screen, Sign Up Screen and Home Screen. Buttons and text actions use route navigation to move between screens.

5. Reusable Widgets

Custom Button: A reusable button component is used for actions such as Login and Create Account.

Custom TextField: A reusable input component supports hints, prefix icons, password visibility and keyboard types.

Product Card: A reusable card displays a product title, category, price and icon and can be reused in both grid and list layouts.

6. Login Screen

The Login Screen uses the reusable TextField and Custom Button widgets. The Sign Up action navigates to the registration screen.

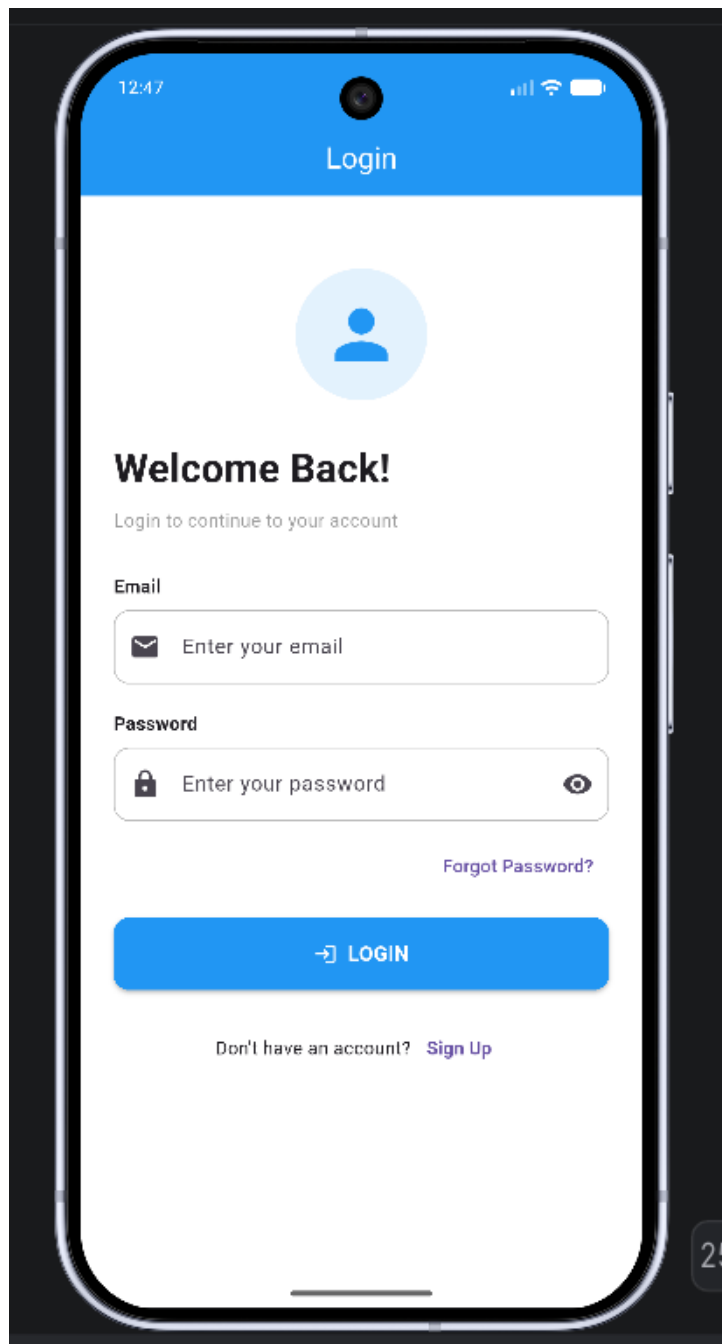


Figure 1 — Login Screen using reusable UI components.

7. GridView Product Layout

The Home Screen displays products in a two-column GridView. Each item uses the reusable Product Card widget.

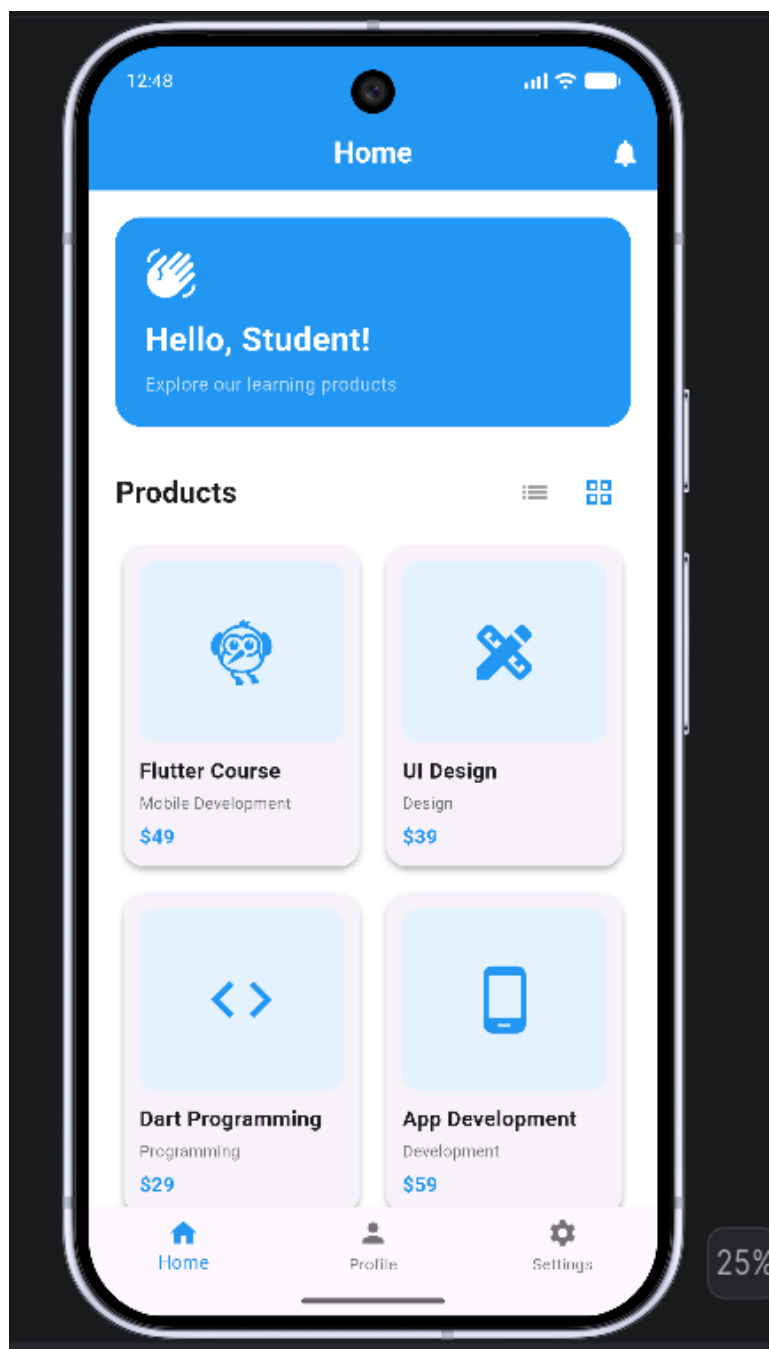


Figure 2 — Product GridView.

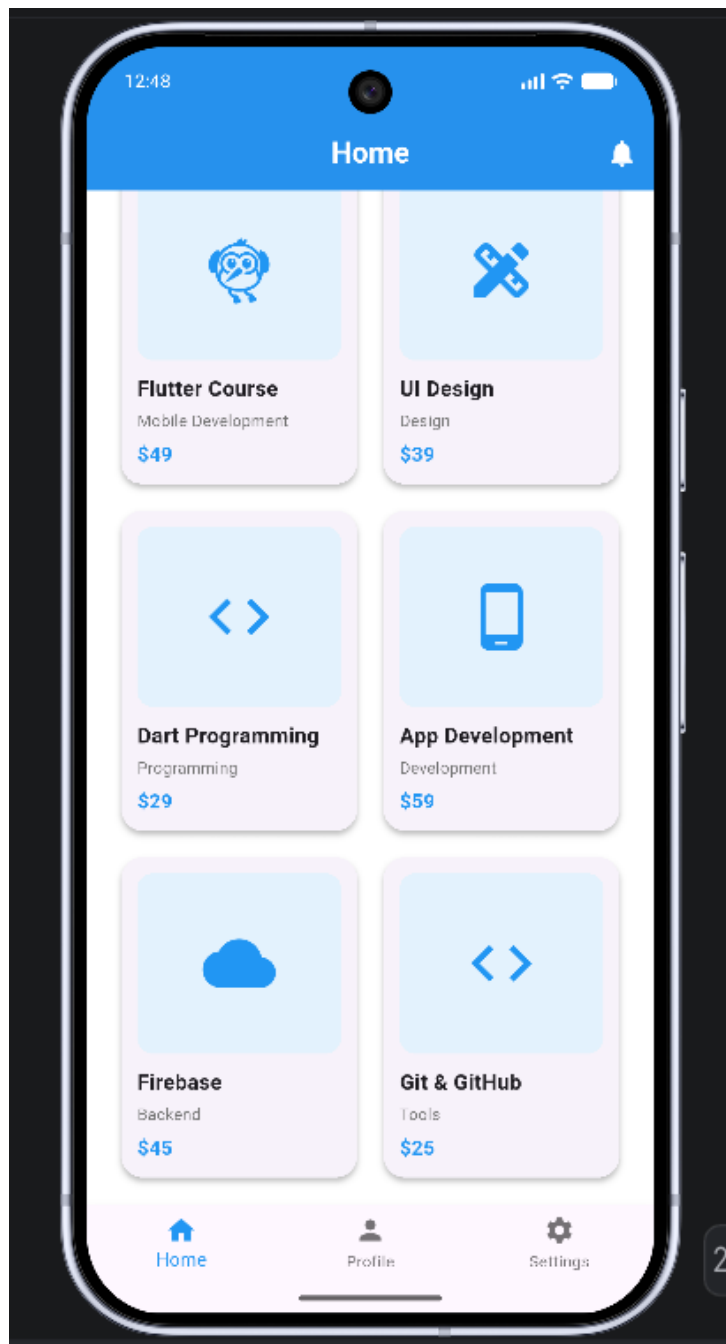


Figure 3 — Full GridView product list.

8. ListView Product Layout

The application also provides a ListView option. The user can switch between grid and list layouts using the layout controls.

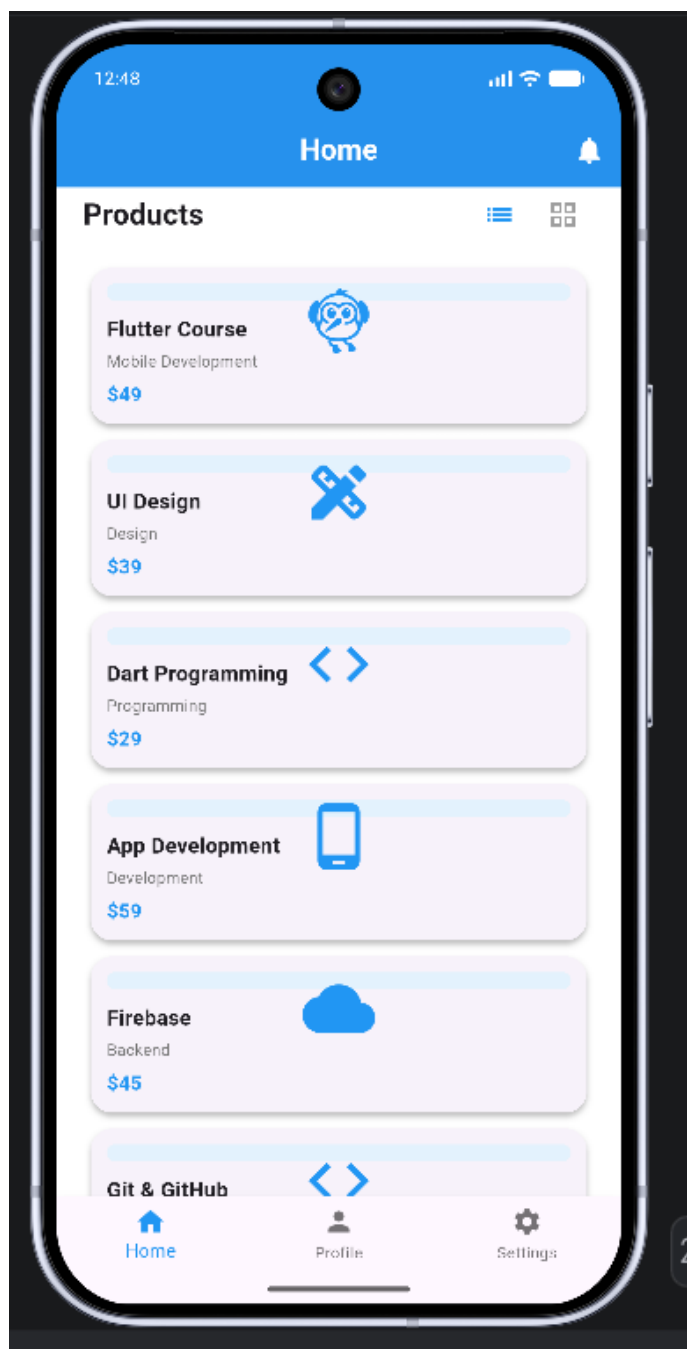


Figure 4 — Product ListView.

9. Organized Project Structure

The project is organized into separate screens and widgets folders. Screen files are kept under screens, while reusable components are kept under widgets. This structure improves code readability, reuse and maintainability.

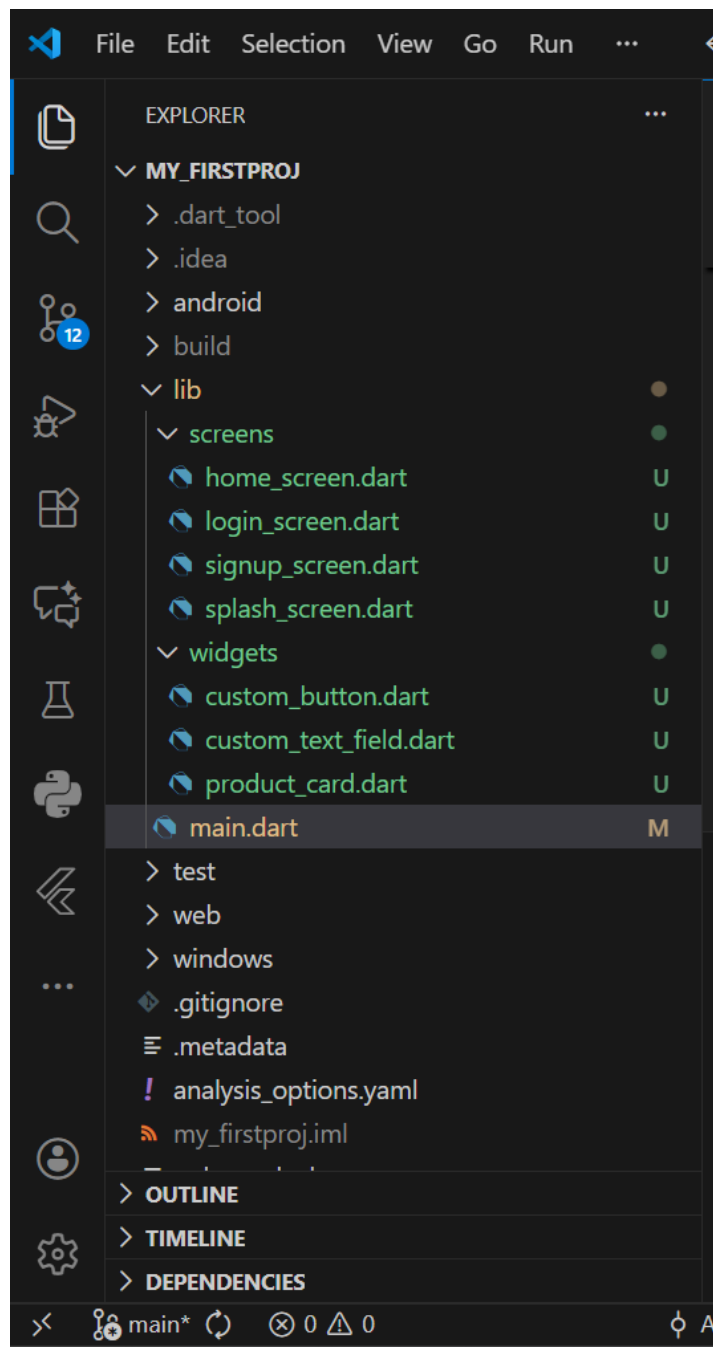


Figure 5 — Organized Flutter project structure in Visual Studio Code.

10. Bottom Navigation Bar

A BottomNavigationBar was added to provide three main sections: Home, Profile and Settings. The selected tab changes the displayed content while keeping the navigation available at the bottom of the application.

11. Conclusion

This task successfully demonstrates Flutter navigation and reusable UI development. The application now has a more organized folder structure, reusable components, multiple product layouts, and bottom navigation. These concepts provide a practical foundation for building larger and more maintainable Flutter applications.